Here we check that the normalization is OK

In[2]:=
```
Integrate[1 / Sqrt[2 Pi t ] (α Exp[- (x - a t ) ^2 / (2 t)] + (1 - α) Exp[- (x - b t ) ^2 / (2 t)]),
 {x, -Infinity, Infinity}]
```

Out[2]= $\sqrt{\frac{1}{t}}\,\sqrt{t}$

Here we check the mean value of the distribution

In[3]:=
```
Integrate[x / Sqrt[2 Pi t ] (α Exp[- (x - a t ) ^2 / (2 t)] + (1 - α) Exp[- (x - b t ) ^2 / (2 t)]),
 {x, -Infinity, Infinity}]
```

Out[3]= $\sqrt{\frac{1}{t}}\,t^{3/2}\,(b + a\,\alpha - b\,\alpha)$

Here we compute the variance

In[4]:=
```
Integrate[(x - t (α a  + (1 - α) b))^2 / Sqrt[2 Pi t ]
  (α Exp[- (x - a t ) ^2 / (2 t)] + (1 - α) Exp[- (x - b t ) ^2 / (2 t)]), {x, -Infinity, Infinity}]
```

Out[4]= $t\,\left(1 - (a - b)^2\,t\,(-1 + \alpha)\,\alpha\right)$ if $\mathrm{Re}[t] \geq 0$

We conclude that for small t, the variance of the distribution is unchanged, and the mean is given by the mean of a and b, as expected.